How Spotify and Netflix Build Their ML Pipelines

Bring your W5S1 worksheet. Today we compare the music recommendation system you designed from scratch against what Spotify and Netflix actually built. Fill this worksheet in as we go through each phase — do not go back and change your W5S1 answers.

Phase 1 (0:00–0:10) — Compare Your Design, Before Any Reveal

Answer honestly from your W5S1 worksheet — what you actually wrote, not what you wish you had written. This is not graded.

1. What algorithm or model did you use for candidate generation?

2. How often did your system retrain?

3. How did you handle a brand-new song with no interaction data?

4. What was your primary evaluation metric?

5. Did you separate candidate generation from ranking?

The Two Systems We Are Studying Today

Spotify Discover Weekly

Scale: 30 million users, every Monday
Output: 30-song personalised playlist
Cadence: Weekly batch job
Key tech: word2vec on playlist data
Cold-start: CNN audio embeddings for new songs

Netflix Recommendations

Scale: 200+ million members
Output: Ranked content rows
Cadence: Candidate model daily, ranker hourly
Key tech: Two-tower neural network + ANN
North star: Streaming hours, not CTR

Today's Session at a Glance

Phase	Time	Activity
1. Compare	0:00–0:10	Answer 5 questions from your W5S1 design, before any reveal
2. Spotify	0:10–0:50	word2vec collaborative filtering, audio CNN cold-start, weekly batch pipeline
3. Netflix	0:50–1:20	Two-tower candidate generation, separate ranker, streaming-hours evaluation
4. Discussion	1:20–1:45	Fill the compare table, share what surprised you
5. Lessons	1:45–2:00	Five takeaways, preview of Week 6 (LLM serving)

Name: _____ Group: _____ Date: _____

Phase 2 (0:10–0:50) — Spotify Discover Weekly Deep-Dive

Fill this in as the instructor walks through the case study pages 2–3.

Model 1 Collaborative Filtering via word2vec on Playlists

The core analogy: playlists are treated as _____, songs are treated as _____.
Songs that co-occur often in playlists end up _____ in embedding space.

Why does this work without any audio analysis? (instructor's answer)

Model 2 Content-Based Filtering via Audio CNN

What problem does this model solve that word2vec cannot?

Model 3 NLP on Text: Cultural Context Embeddings

What signal does this capture that the other two models miss?

Spotify's Freshness Trade-off

Q: Why does Discover Weekly update every Monday, and not every hour like Netflix?

Follow-up: What product assumption is embedded in a weekly cadence?

Cold-Start: How Spotify Handles a Brand-New Song

Song age	Available signal	Embedding source
0–7 days		
7–30 days		
30+ days		

Compare to your W5S1 answer (Q3, page 1) — did you handle cold-start explicitly?

Phase 3 (0:50–1:20) — Netflix Two-Tower + Ranking

Fill this in as the instructor walks through the case study pages 4–5.

Stage 1 Candidate Generation: Two-Tower Neural Network

Sketch or label the two-tower diagram: user tower, item tower, shared embedding space, ANN index.

Draw the two-tower architecture here: user features → user tower, item features → item tower, dot product, ANN search → top-1000.

Why can item embeddings be pre-computed offline but not user embeddings?

Stage 2 Ranking: Separate Neural Network with Rich Features

Why does Netflix need a second model instead of ranking directly with the two-tower scores?

What three things does the ranker optimise for simultaneously?

The Freshness Contrast: Spotify vs. Netflix

	Candidate model	Ranker
Spotify		N/A (one-stage)
Netflix		

Why hourly for Netflix's ranker specifically? What changes within a day that the daily candidate model cannot capture?

Why Streaming Hours, Not CTR?

Q: What happens to a model that optimises for CTR instead of streaming hours?

Your W5S1 evaluation metric (from page 1, Q4) — is it an offline metric or an online, business metric like Netflix's?

Phase 4 (1:20–1:45) — Compare Your W5S1 Design: Five Dimensions

Now fill in the "Your design" column honestly. Where did you match production? Where did you diverge?

Dimension	Spotify	Netflix	Your design
Model architecture	word2vec on playlists + audio CNN	Two-tower neural network	
Serving strategy	Weekly batch, pre-computed	Candidate daily, ranker hourly	
Cold-start	Audio CNN for new songs	Feature engineering (recency, genre)	
Evaluation north star	Implicit — stream completion rates	Streaming hours (not CTR)	
Scale of freshness	Weekly update cycle	Hourly ranker retraining	

One thing that surprised me most about how Spotify or Netflix actually built this:

One thing I got right in my W5S1 design that matches production:

One thing I over-engineered relative to what Spotify or Netflix actually built:

Phase 5 (1:45–2:00) — Five Lessons to Take Away

1. **Production uses proven architectures, not the latest research.** word2vec (2013) and two-tower (2016) still run at scale in 2024.
2. **Freshness strategy is a first-class architectural decision.** Weekly vs. hourly determines infrastructure, cost, and product experience.
3. **Evaluation design determines what your model actually optimises.** CTR as a north star produces clickbait.
4. **Cold-start is a product problem, not an ML afterthought.** Spotify built a dedicated pipeline for it.
5. **Scale forces separation of concerns.** At 200M users, one model cannot do everything.

Which of these five lessons applies most directly to a gap in my own W5S1 design, and why?

Looking Ahead: Week 6 — Serving Large Language Models

Same structure next week: Session 1 you design the system, Session 2 we reveal what OpenAI and Cohere actually built.

LLM inference is fundamentally different from recommendation serving. What do you think changes for: what is stateful vs. stateless, what is pre-computable vs. not, and the latency target for a conversational system?
